

FINAL REPORT

1) Back testing the rule for 5-year period:

Location	Loss Year	Total Months	Total Firings	During Loss Year	Outside Loss Year
Healthy Forest	None	53	6	0	6
Dense Forest 1	2022	56	9	2	7
Dense Forest 2	2020	54	24	4	20
Medium Forest	2022	52	10	3	7
Sparse Forest	2020	57	12	3	9
Very Sparse Forest	2020	55	14	3	11

2) Residual NDVI Range:

Considered parameters:

Pre-disturbance years only:

- Dense Forest 1: 2019-2021 (since loss is in 2022)
- Dense Forest 2: 2019 (since loss is in 2020)
- Medium Forest: 2019-2021 (since loss is in 2022)
- Sparse Forest: 2019 (since loss is in 2020)
- Very Sparse Forest: 2019 (since loss is in 2020)

Growing season only: May-September

Pixel	Pre-loss Years Used	Min Residual	Max Residual	Mean Residual	Std. Dev.
Dense Forest 1	2019–2021	-0.1727	-0.0582	-0.1084	0.0307
Dense Forest 2	2019	-0.2146	-0.1484	-0.1832	0.0249
Medium Forest	2019–2021	-0.1409	-0.0203	-0.0813	0.0298
Sparse Forest	2019	-0.2664	-0.0367	-0.1001	0.0938
Very Sparse Forest	2019	-0.3044	-0.1737	-0.2074	0.0550

Minimum NDVI Threshold:

Pixel	NDVI Threshold
Healthy Forest	-0.1176
Dense Forest 1	-0.1727
Dense Forest 2	-0.2146
Medium Forest	-0.1409
Sparse Forest	-0.2664
Very Sparse Forest	-0.3044

3) Flagged Disturbance Window:

Here, the flagged disturbance window is defined as the entire continuous sequence of months where the NDVI residual remains below the pixel-specific threshold after the rule first triggers.

The resulting windows per pixel:

Pixel	Flagged Disturbance Window
Healthy Forest	No qualifying disturbance window
Dense Forest 1	May 2022 - Oct 2022
Dense Forest 2	Apr 2020 - Jul 2020
Medium Forest	No qualifying disturbance window
Sparse Forest	No qualifying disturbance window
Very Sparse Forest	May 2020 - Jul 2020

4) Baseline VH:

The mean baseline VH is computed over:

- pre-disturbance years
- growing season only

Pixel	Baseline Mean VH (dB)
Dense Forest 1	-14.90
Dense Forest 2	-18.08
Medium Forest	-13.84
Sparse Forest	-14.90
Very Sparse Forest	-14.59

5) Mean VH During the Flagged Window:

Pixel	Flagged Window Mean VH
Dense Forest 1	-13.55 dB
Dense Forest 2	-18.06 dB
Very Sparse Forest	-14.48 dB

6) VH Difference:

$$\Delta VH = VH_{\text{flagged}} - VH_{\text{baseline}}$$

Pixel	Baseline VH	Flagged VH	ΔVH (dB)
Dense Forest 1	-14.90	-13.55	+1.3433
Dense Forest 2	-18.08	-18.06	+0.0238
Very Sparse Forest	-14.59	-14.48	+0.1075

* ΔVH is positive the radar signal did not drop at all it actually increased or stayed flat.

A real clearing/deforestation event requires a negative change / drop in radar signal.

Tested VH threshold	Pixels satisfying threshold
≥ 0.25 dB drop	0 / 3
≥ 0.50 dB drop	0 / 3
≥ 1.00 dB drop	0 / 3

*Quantifying VH demonstrates that Sentinel-1 does not consistently exhibit a measurable backscatter decrease during the detected disturbance windows.

7) Confusion Matrix:

- **True Positive (TP):** A loss pixel correctly identified as a clearing event by the detection rule.
- **False Positive (FP):** A non-clearing (stable) pixel incorrectly identified as a clearing event.
- **True Negative (TN):** A non-clearing (stable) pixel correctly identified as a non-clearing event.
- **False Negative (FN):** A loss pixel incorrectly classified as a non-clearing event because the rule failed to detect the disturbance.

	Actual Clearing	Actual No Clearing
Rule = Clearing	0 (TP)	0 (FP)
Rule = No Clearing	5 (FN)	1 (TN)

- **Healthy Forest (TN):** Correctly classified as non-clearing because its NDVI residuals never breached its historical pre-loss minimum threshold.
- **Dense Forest 1 (FN):** Missed because despite breaching the NDVI threshold, its radar signal actually *increased* by +1.34 dB instead of dropping.
- **Dense Forest 2 (FN):** Missed because during its optical disturbance window, its radar signal remained virtually flat (+0.02 dB change), failing the required radar drop threshold.
- **Medium Forest (FN):** Missed at the first step because its optical NDVI residuals never dropped below its strict historical pre-loss minimum for two consecutive months.
- **Sparse Forest (FN):** Missed at the first step because its optical NDVI residuals failed to continuously breach its strict historical pre-loss minimum threshold.
- **Very Sparse Forest (FN):** Missed because its radar backscatter remained flat (+0.11 dB change) during the flagged optical drop window, failing to confirm structural loss.

8) Conclusion:

The revised detection rule successfully eliminated the false positives observed in the original implementation by introducing a pixel-specific NDVI magnitude threshold and a quantitative Sentinel-1 VH backscatter criterion. The Stable Forest control pixel was correctly classified as a non-clearing event, demonstrating that the revised thresholds effectively suppress detections caused by normal seasonal variability and radar noise.

However, when evaluated across the five known loss pixels, none of the clearing events satisfied both the revised NDVI and VH conditions simultaneously. Consequently, all five clearing pixels were classified as non-clearing events, resulting in five false negatives and no true positives.

This outcome indicates that, although the revised rule is highly conservative and substantially improves specificity by avoiding false alarms, it sacrifices sensitivity and fails to detect genuine forest-clearing events. The VH backscatter changes associated with the observed disturbances were generally smaller than expected or too variable to satisfy the quantitative threshold, despite clear evidence of vegetation loss from the NDVI residuals.

END OF REPORT